

Syed Muhammad Abbas

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Professional Summary

Software Engineer and NUST CS senior with hands-on experience building scalable backend systems, production-grade REST APIs, and AI-powered pipelines. Passionate about Clean Code, SOLID design, and OOP; eager to contribute to a cross-functional agile team and grow through rigorous engineering standards.

Technical Skills

Languages: Java, Python, TypeScript, SQL, C++, C#, Bash

Backend & Databases: Spring Boot (MVC, DI, JPA/Hibernate), REST APIs, Microservice Architecture, FastAPI, PostgreSQL, Node.js

AI & Automation: n8n, LangChain, LLMs, Generative AI, OpenCV, YOLO, MediaPipe

Cloud, DevOps & Frontend: AWS, Docker, CI/CD, Next.js, React, Git, Postman

Core Concepts: OOP, SOLID Principles, Clean Code, Data Structures & Algorithms, Distributed Systems

Professional Experience

Software Engineer Intern, Horizon Tech Services – Islamabad, PK

June 2025 – Aug 2025

- Engineered robust REST APIs using FastAPI for a real-time data processing system, optimizing endpoints to achieve sub-100ms latency and reliable concurrent communication.
- Adopted AI inference pipelines (MediaPipe, TFLite) for real-time gesture recognition on Android edge devices, bridging backend infrastructure with mobile constraints.

JST Research Scholar, Japan Science & Technology Agency – Tokyo, JP

July 2019

- Competitively selected as a top national scholar to conduct robotics research at Tokyo Institute of Technology; applied analytical thinking in research-to-production workshops with leading international scientists.

Selected Projects

Gasthaus – Full-Stack Enterprise Restaurant System

github.com/sm-abbas19

- Architected an enterprise backend using **Java** and **Spring Boot (MVC, JPA/Hibernate)**, applying a layered Controller/Service/Repository architecture, SOLID principles, and dependency injection to build 10+ scalable **REST APIs** with DTO-based API contracts and Spring validation.
- Designed a normalized **PostgreSQL** schema with index optimization and transaction management, ensuring data integrity across 100+ daily simulated operations.
- Implemented a **STOMP over WebSocket** pub/sub broker with global exception handling, achieving sub-50ms state synchronization across 3 role-based clients; integrated Gemini API for automated business analytics.

SecureVision (FYP) – Real-Time AI Surveillance Pipeline

github.com/sm-abbas19/SecureVision

- Architecting an end-to-end pipeline using YOLOv11 and BOTSORT for real-time threat detection with a React/Next.js monitoring dashboard.
- Developing a custom fusion layer integrating SLMs for context verification, reducing false positives by 40% through QA and data validation.
- Optimizing inference via multi-threading strategies targeting 25+ FPS on consumer hardware; mentoring 2 team members on Computer Vision workflows and documentation.

PCA-Praxis – Secure Agent Execution Sandbox

github.com/sm-abbas19/PCA-Praxis

- Orchestrated an autonomous AI agent (Phi-3 SLM) translating natural language into Linux shell commands, executed inside an ephemeral Docker sandbox with strict security isolation and resource management.

Education

NUST (SEECS), Pakistan – BS in Computer Science (CGPA: 3.33/4.00)

2022 – Present

Coursework: Data Structures & Algorithms, Relational Database Design, Distributed Computing, Operating Systems, Web Engineering, Machine Learning

Languages and Certifications

Languages: English (Fluent), Urdu (Native), German (Elementary - A1/A2) **Meta:** Back-End Developer Professional Certificate

DeepLearning.AI: Deep Learning & Machine Learning Specializations **AWS/DeepLearning.AI:** Generative AI with Large Language Models